

Problem Set 4 - Control Flow: Selection

Hello students,

the goal of this problem-set is to familiarize with the different control flow statements, in particular with selection statements.

During the last lectures you also learned about the use of the string formatter, from this problem-set you are expected to use it.

An example could be:

```
answer = 42
print(f"The meaning of life is {answer}!")
```

Which prints: The meaning of life is 42!

Assignment 4.1 - Properties of the numbers (optional)

Write a program that asks the user for two positive integer numbers and that determines:

- If the first number is a multiple of the second, print the result of the division, otherwise a message stating they are not multiple;
- Print if the first number is smaller, bigger or equal in respect to the second;
- If the difference between the first number and the second is bigger than the difference between the second and the first or vice-versa.

Here a few examples of the expected output (in bold red the input by the user):

```
Insert the first number: 4
Insert the second number: 2

The result of the division is: 2
4 is bigger than 2
The difference (4 - 2) is bigger than (2 - 4)
```

```
Insert the first number: 2
Insert the second number: 7

The two numbers are not multiple
2 is smaller than 7
The difference (2 - 7) is smaller than (7 - 2)
```

```
Insert the first number: 15
Insert the second number: 15

The result of the division is: 1
The two numbers are equal!
The difference between the two numbers is equal!
```

Assignment 4.2 - Range numbers (optional)

Write a program that asks the user for one positive integer number between 80 and 189 (both included). If the number is not between the specified range, the program must print an error message, otherwise it should print in which range it is: between 80 and 99, 100 and 119, ... , 160 and 189.

The output must have the exact same format as this examples (in bold red you can see the user inserted values):

```
Insert a number between <80> and <189>: 142
The number [142] is in the range between <140> and <159>.
```

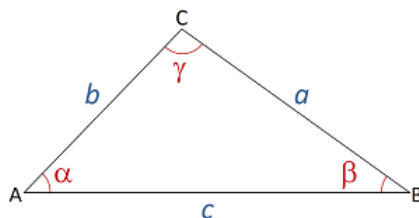
```
Insert a number between <80> and <189>: -5
The number [-5] is not in the range between <80> and <189>.
```

Assignment 4.3 - Law of cosines (optional)

Write a program that, using the law of cosines, calculate the length of a side of a triangle, having only the two other sides and the angle between them in degrees (between 0 and 180).

The program must first ask which sides will be inserted (b and c, a and c or a and b), get their values and the angle between them from the user, and finally calculate and show the missing side name and size.

If non-valid values are inserted, you are expected to print an error message.



$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$

$$b^2 = a^2 + c^2 - 2ac \cos \beta$$

$$c^2 = a^2 + b^2 - 2ab \cos \gamma$$

Hint 1: Use the python standard library `math` module to do the calculations.

```
from math import radians, sqrt, cos
```

Hint 2: The angles measurements (α , β , and γ) must be converted to radians to be used in the formula.

Assignment 4.4 - Engines management (optional)

You are an engineer at NASA (Not Another Supsi Aircraft) and you need to write a piece of firmware (code) to manage the calibration of the engines of the new Falcon S (where "S" stays for SUPSI).

These aircrafts have different engine-configurations depending on the field of application.

Your job is to write a software that is able to tell the maximum pressure the engines can hold (it is a steam-engine for some reason).

This calculation is based on the engine-type and the length of the aircraft.

The informations you got from the engineering team are:

- If the engine is of type 1 and the length of the aircraft is greater than 20m, the maximum pressure is 200 bar.
- If the engine is of type 2 and the length is between 10m (excluded) and 20m (included), or the engine is of type 1 and the length is less than 20m, the maximum pressure is 150 bar.
- If the engine is of type 3 or the length is less than 10m, the maximum pressure is 100 bar.
- Otherwise 50 is safe.

The program should work like this:

```
Insert the engine's type: 1
Insert the aircraft's length: 15
>>> The maximum pressure is 150 bar.
```

```
Insert the engine's type: 5
Insert the aircraft's length: 30
The maximum pressure is 50 bar.
```

Assignment 4.5 - Body mass index (optional)

Body mass index (BMI) is a measurement used to estimate whether a person is at a healthy weight. This is a generic indicator not taking into consideration many factors such as activity level and lean body mass.

The formula is:

$$bmi = \frac{weight(kg)}{[height(m)]^2}$$

Alternatively you can use the height in centimeters:

$$bmi = \frac{weight(kg)}{[height(cm)]^2} * 10000$$

The result is a number representing the weight ranges, the ranges are as follows:

- Less than 18.6 : Underweight
- Between 18.5 and 24.9 : Healthy weight
- Between 25 and 29.9 : Overweight
- More than 30 : Obese

Again, keep in mind this is not an accurate measurement! Don't stress it!

Write a program asking for the user's height and weight. Calculate the BMI. Print what the health range is (from the aforementioned list).

Example output:

```
Enter your height (cm): 170
Enter your weight (kg): 65

Your BMI is: 22.49
You are in a Healthy Weight range!
```